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Learning to cooperate against ensembles of diverse opponents

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Abstract
The emergence of cooperation in decentralized multi-agent systems is challenging; naive implementations of learning algorithms typically fail to converge or converge to equilibria without cooperation. Opponent modeling techniques, combined with reinforcement learning, have been successful in promoting cooperation, but face challenges when other agents are plentiful or anonymous. We envision environments in which agents face a sequence of interactions with different and heterogeneous agents. Inspired by models of evolutionary game theory, we introduce RL agents that forgo explicit modeling of others. Instead, they augment their reward signal by considering how to best respond to others assumed to be rational against their own strategy. This technique not only scales well in environments with many agents, but can also outperform opponent modeling techniques across a range of cooperation games. Agents that use the algorithm we propose can successfully maintain and establish cooperation when playing against an ensemble of diverse agents. This finding is robust across different kinds of games and can also be shown not to disadvantage agents in purely competitive interactions. While cooperation in pairwise settings is foundational, interactions across large groups of diverse agents are likely to be the norm in future applications where cooperation is an emergent property of agent design, rather than a design goal at the system level. The algorithm we propose here is a simple and scalable step in this direction.

Keywords Cooperation · Iterated Prisoner's dilemma · Evolutionary game theory · Best response · Population games

1 Introduction

Cooperation is an essential feature of human social interactions and a desirable feature in multi-agent systems. Cooperation happens when agents pay a cost in order to help others. Not cooperating is typically a dominant strategy, but other equilibria may also exist where agents can mutually benefit from learning to work together. To be successful, artificially intelligent agents need the ability to cooperate with humans [1] and other machines in diverse environments [2–4]. These environments can encompass interactions that range from fully cooperative—where the incentives of all parties are aligned—to mixed-motive games where agents benefit from cooperation, to fully competitive games. Most games of cooperation sit between these two extremes. Convergence to undesirable equilibria is common in decentralized settings [5–7].

In games of cooperation, self-interested agents need to be competitive by avoiding exploitation, but they also need to have the ability to reap the gains to be had by working together whenever these synergies are available. This tension between individual rewards and group outcomes is the defining feature of cooperation [8].

Reinforcement learning (RL) has achieved unparalleled success in fully cooperative and fully competitive multi-agent environments. However, the application of RL to general-sum games played by many autonomous agents is not trivial. In fully decentralized Multi-agent Reinforcement Learning (MARL) where agents learn simultaneously and without a central controller, from the perspective of one agent, every other agent is part of their environment. Here, non-stationarity makes it difficult for naive reward maximizing agents to converge to equilibria even with simple reward structures [9].

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In this paper, we study decentralized cooperation. We aim for environments in which agents face a series of interactions with diverse opponents. Examples include self-driving cars, which would resolve a number of interactions as part of a route or sequential transactions on a market floor. While opponent modeling may be successful when an agent learns by playing against another, sequences of interactions with several diverse agents may render this approach infeasible. We thus focus on formulating an intrinsic reward function that scales well in this kind of environment, fostering cooperation when there is an opportunity to do so, while being resilient to play against agents that do not cooperate or do not use the same reward function.

To be more precise, we consider an environment characterized by the following conditions:

1. A population of decentralized agents playing a two-player repeated game (with well-mixed random matching).
2. Each agent has access to the world model dynamics (states, actions, rewards, and transition probabilities).
3. Agents have no access to the (possible) internal rewards, states, or policies of their opponents.

MARL has been successful in addressing cooperation through opponent modeling [6, 10] and centralized rewards [11]. Opponent modeling attempts to identify the opponent's strategy and actively respond to it, making it easier to achieve cooperation. Centralizing rewards refers to agents attempting to improve the collective outcome instead of pursuing their individual rewards. While centralization is a simple approach to achieve cooperation, the success of centralized architectures relies on the ability to alter the objective functions of individual agents. This assumption is hard to match in a realistic population setting where exchanging rewards may not be possible and communication is limited.

Opponent modeling on the other hand can work in fully decentralized environments, but its computational cost may have a strong impact in large environments where it is necessary to keep track of all others. Moreover, in settings with a diverse ensemble of agents, individuals may only ever get to interact with a handful of others and agents are often anonymous. This, in principle, can also affect the observability necessary to successfully model opponents.

In this paper, we introduce best response guided agents (BRG), which consider an intrinsic reward that measures their ability to make a reward maximizing opponent exploitable in the future. This is computed on the basis of information that the agent already has about his own policy and the game itself. This intrinsic reward allows them to successfully navigate the reward structure in cooperation problems, even when the environment features large ensembles of diverse agents. Our agents learn to adopt reciprocating strategies that reward cooperation and penalize defection, in turn guiding other agents toward desirable equilibria.

Agents can either use their knowledge of the game or use self-play [12, 13] to calculate this intrinsic reward. The proposed method does not require specific knowledge or inferences over the strategies of others, which makes it scalable in population games. In addition, preserving the self-centered nature of individual agents makes them less prone to exploitation while maintaining cooperation.

We test our approach on a range of different cooperation games. We start with a standard Iterated Prisoner's Dilemma (IPD) [14], a game with infinitely many equilibria [15] where standard agent-centric RL algorithms are known to fail to converge to cooperation [6, 16]. We then test our approach on a stochastic game [17], and a standard board game where cooperation and defection are no longer elementary actions, but properties of policies [18]. We also verify that the approach is robust even if the strategic interactions do not present cooperation opportunities, including fully competitive games.

The rest of this paper is organized as follows. In Sect. 2, we review the relevant literature on MARL, cooperation, and population games. Our method is introduced in Sect. 3 and is empirically tested across different environments in Sects. 4–6. Section 7 concludes the paper and discusses extensions and limitations.

2 Related Work

Interest in Cooperative AI has led to diverse areas of research such as communication [19–21], social preferences [9], and collective decision making [22–24]. Most of this research focuses on scenarios where agents have fully aligned interests [25–27]. However, when agents interacting together cannot be assumed to have aligned interests, these types of solutions cannot be applied safely. Without a central control of individual agents, self-interested agents have to strategize.

Dynamics of strategic interaction between self-interested agents have long been studied in the field of game theory. Game theory deals with the analysis of equilibrium strategies of fully rational players in static games that are fully known to the players. In this context, a Nash equilibrium solution tells us what is an expected outcome, such as the famous result that only mutual defection makes sense in a prisoner's dilemma problem.

However, many real-world situations are not single-shot games. Most board games people play are executed over multiple time steps with evolving conditions (e.g., pieces in play, and their positions, in chess) and also in the political

arena the same people interact continuously. Such situations have been formalized as *repeated* games and as *Markov* games. The former are games from game theory but with repeated interactions resulting in complex strategy spaces, while the latter are games with complex state dynamics that depend on the joint actions of players. The two branches give rise to two different but interconnected research areas.

A repeated game raises the question how the notion of Nash equilibrium changes when agents are aware that they will be playing multiple rounds in succession. It has been shown that in the Iterated Prisoner's dilemma game [14], infinitely many equilibria exist, including socially efficient equilibria that sustain cooperation, as well as those where defection is prevalent [28]. Evolutionary Game Theory (EGT) explores the dynamics of agent strategies over time in such a complex strategy space and how stable strategies may emerge if agents have to play in a population of other evolving agents [29, 30].

In Markov games, one of the central challenges is reasoning about how the current decision affects the future state of the world and in turn the strategic consequences of that outcome. As such, work in this space deals with learning to choose appropriate actions, for example, through reinforcement learning or Monte Carlo tree search, which is successful in zero-sum games [31–33]. However, to learn optimal actions, we necessarily have to reason about the strategic opponent, and as such learning a strategy in multi-agent reinforcement learning (MARL) has a deep connection with EGT, as shown by Tuyls and Nowé [34]. As such, repeated games such as IPD are common test beds in other studies of MARL, e.g., [6, 35, 36].

Our work is focused on learning strategies in games of cooperation where individuals are competing, but have incentives to reap higher benefits when working together. The key to achieving cooperation in this setting is reciprocity. Both, Eccles et al. [37] and Lerer and Peysakhovich [18] use reciprocity to design algorithms that achieve better outcomes in general-sum games with RL. These algorithms use a combination of purely cooperative and purely competitive strategies to enforce reciprocity. In contrast, our BRG agents do not explicitly prescribe reciprocity to achieve cooperation but naturally tend to achieve it following the intrinsic reward.

Another class of algorithms which includes WoLF [38], JAL [12], AWESOME [13], and Lanctot et al [39] use best response and self-play in general-sum games. WoLF proposes using varying learning rates to achieve convergence and provide proof for a subset of iterative matrix games. JAL attempts to understand the value of joint actions and requires maintaining beliefs about others strategies. Claus and Boutilier [12] show that JAL does not always converge to the optimal equilibrium in situations with many equilibria. AWESOME uses a pre-calculated Nash equilibrium of the one-shot game to guarantee convergence. Hu and Wellman [10] introduce the Nash-Q algorithm where agents use Nash equilibria calculations with opponent modeling to reach higher payoffs in general-sum games. Nash-Q operates with the assumption that opponents attempt to reach the calculated equilibrium and convergence varies with the type of Nash equilibrium chosen. These algorithms focus on convergence more than on achieving Pareto-optimal equilibria or shaping opponents behavior for better future rewards.

Numerous other approaches to solving MDPs in multi-agent settings involve accounting for others' beliefs based on cognitive hierarchies such as level-k solution concept [40–42]. These methods lead to more cognitively advanced agents capable of teaching and learning from other agents [43, 44]. While our agents do not explicitly account for distinct hierarchical levels, the concept of using information from best responses to naïve learners shares some similarities with these approaches. Our agents are not modeling others explicitly, but assuming they are responding reasonably to the strategy of our learning agent.

Although our work does not focus on purely competitive settings or include opponent modeling, the proactive approach of BRG agents to exploit opponents, in the long run, relates to existing research on opponent modeling and lookahead type algorithms [45, 46]. We use these as inspiration to identify an intrinsic reward based on best response and lookahead calculations. Most of these methods assume a limited number of agents in a controlled environment. It is therefore difficult to apply them to environments with ensembles of diverse agents.

We assume agents are matched for playing a game in each round, drawn from a population. The reward is taken to be the average reward over a series of matches. Importantly, in this scenario agents are anonymous, self-interested and there is no central control or institution mediating interactions.

3 Method

The naïve form of decentralized MARL considers opponents to be a part of the agent's environment and ignores the non-stationary nature of the environment in multi-agent systems [47]. As a result, naïve learners lose the ability to proactively explore the combined action space, tending to

converge to undesirable equilibria in mixed-motive multi-agent settings. If agents can account for how their current policy affects their future rewards, they could possibly avoid this mishap.

BRG agents achieve this by assuming a best responding opponent with perfect information. A measure of the impact of the current policy on future rewards is then given by the reward of the best counter-response to the best responding opponent. This becomes the intrinsic reward of a BRG agent. Instead of only maximizing the current reward, BRG agents attempt to maximize the current reward plus the intrinsic reward. We discuss this in detail in the following section.

3.1 Preliminaries

When evaluating a best response, we construct a single-agent Markov Decision Process (MDP) from the repeated game. $G = \langle \mathcal{S}, \mathcal{A}, \mathcal{T}, \mathcal{R}, \gamma \rangle$ where $s \in \mathcal{S}$ is the state of the environment defined using memory, $a \in \mathcal{A}$ is the action taken by the focal agent, $\mathcal{R}$ is the reward function specifying the immediate rewards as given by the payoff matrix, $\mathcal{T}$ is the state transition matrix, and $\gamma \in (0, 1)$ is the discount factor of future rewards. The probability of taking each action in a given state is specified by the policy of an agent. The state transition matrix uses the opponent policy's action selection probabilities and therefore assumes a fixed opponent policy for the duration of a game.

To identify a best response against a fixed strategy, we use Q-value iteration. The recursive Bellman equation [48] defines the expected value of a state, $V(s)$, and the expected value of taking action a in state s and acting optimally from then onwards, $Q(s, a)$. This quantity is calculated iteratively, starting from $Q_0(s, a) = 0$, as

$$Q_{k+1}(s,a) = \sum_{s' \in \mathcal{S}} \mathcal{T}(s,a,s') \left[\mathcal{R}(s,a,s') + \gamma \max_{a' \in \mathcal{A}} Q_k(s',a') \right] \quad \forall s, a \tag{1}$$

Once the state values converge, the maximum value action in a given state s is followed to identify the optimal action sequence.

For learning, our agents use policy gradient [49] and update their policy, π, by performing gradient ascent on the expected discounted reward with respect to the policy parameters, $\theta(s, a)$. We use Softmax for the policy parametrization, meaning

$$\pi_\theta(s,a) = \frac{\exp(\phi(s,a) \cdot \theta)}{\sum_{k=1}^{|\mathcal{A}|} \exp(\phi(s,a_k) \cdot \theta)}, \tag{2}$$

where $\phi(s, a)$ is the feature vector related to a state, action pair.

3.2 Intrinsic reward calculation

The intrinsic reward calculation for the state–action pair, (s', a'), under policy, π, aims to measure how beneficial taking action a' in state s' at this point will be in future episodes (see Algorithm:1, lines 4–11). To compute the intrinsic reward for (s', a'), we assume the focal agent deterministically picks action a' in state s', while following the current policy π_i in all other states. Thus, FIX_POLICY on line 6 computes a modified policy $\pi_i^{s',a'}$ for which

$$\pi_i^{s',a'}(s,a) = \begin{cases} 1, & \text{if } s = s', a = a', \\ 0, & \text{if } s = s', a \neq a', \\ \pi_i(s,a), & \text{otherwise.} \end{cases} \tag{3}$$

As the next step to identifying future rewards, we assume the opponent will be best responding and calculate the best response, $\bar{\pi}_i$, against $\pi_i^{s',a'}$ using value iteration, Algorithm:1, line 7.

$$Q^{\text{BR}}, \bar{\pi}_i \leftarrow \text{BEST_RESPONSE}(\pi_i^{s',a'}) \tag{4}$$

Here, Q^{BR} refers to the Q-values, the expected reward of state–action pairs at convergence in value iteration.

Assuming that the opponent will eventually learn to follow this best response policy $\bar{\pi}_i$, we preempt their best response by computing the focal agent's best response to this future policy on line 8 of Algorithm 1,

$$Q^{\text{BR}^2}, \bar{\bar{\pi}}_i \leftarrow \text{BEST_RESPONSE}(\bar{\pi}_i) \tag{5}$$

The value of each action when best responding is given by Q^{BR^2}. The optimal *preemptive* action to follow in state s' according to this best response could be different from the *intended* action a', i.e., $a' \neq \text{argmax}_a(Q^{\text{BR}^2}[s', a])$. Therefore, our intrinsic reward of the state–action pair (s', a') is given by the value of the optimal action to take according to Q^{BR^2},

$$\text{IR}[s', a'] \leftarrow \max_a \left(Q^{\text{BR}^2}[s', a] \right) \tag{6}$$

It is important to note here that generally, $\text{IR}[s', a_j] \neq \text{IR}[s', a_k]$ for $j \neq k$, because the input policy $\pi_i^{s',a'}$ depends on a'.

Algorithm 1 BRG agent i, training on population game G

1: $\pi_i \leftarrow$ initial policy, $ep \leftarrow 0$
2: **while** $ep <$ limit **do**
3: $\quad exp \leftarrow \text{PLAY_GAMES}(G, \pi_i, \text{batch size})$ $\quad \triangleright\ exp = [\langle s_1, a_1, r_1 \rangle, ...]$
4: $\quad \text{IR}[s, a] \leftarrow 0,\ \forall s, a$
5: $\quad$ **for** $s' \in S, a' \in A$ **do**
6: $\quad\quad \pi_i^{s',a'} \leftarrow \text{FIX_POLICY}(\pi_i, s', a')$ $\quad \triangleright$ Equation (3)
7: $\quad\quad Q^{\text{BR}}, \bar{\pi}_i \leftarrow \text{BEST_RESPONSE}(\pi_i^{s',a'})$
8: $\quad\quad Q^{\text{BR}^2}, \bar{\bar{\pi}}_i \leftarrow \text{BEST_RESPONSE}(\bar{\pi}_i)$
9: $\quad\quad \text{IR}[s', a'] \leftarrow \max_a \left(Q^{\text{BR}^2}[s', a]\right)$
10: $\quad$ **end for**
11: $\quad exp_{\text{IR}} \leftarrow \eta r_j + (1 - \eta)\frac{\text{IR}[s_j, a_j]}{n},$ $\quad \forall \langle s_j, a_j, r_j \rangle \in exp$
12: $\quad \pi_i \leftarrow \text{POLICY_GRADIENT}(\pi_i, exp_{\text{IR}})$
13: $\quad ep \leftarrow ep + 1$
14: **end while**

The Q-values capture the expected reward of taking an action in a given state for an entire episode and are not on the same scale as environmental rewards r. Therefore, we scale the Q-value by dividing it by the expected number of steps per episode, n. For repeated games with continuation probability, δ, n is expected to be $1/(1-\delta)$. Putting it all together, we obtain the new utility function,

$$\mathcal{U} = \eta r + (1 - \eta)\frac{\text{IR}[s_j, a_j]}{n}, \tag{7}$$

where $\eta \in [0, 1]$ specifies the extent to which the agent should focus current and future rewards. See Algorithm:1, line 11. To achieve best performance η needs to be sufficiently low at the start of learning to allow for adequate exploration.

To compute the intrinsic reward for all state–action pairs, we need to compute $2|\mathcal{S}||\mathcal{A}|$ best response value iteration calls, which are themselves polynomial in the size of the single-player MDP folding the target policy into transition function $\mathcal{T}$.

- Space complexity: Each value iteration is run sequentially, thus memory for it can be reused. Therefore, the intrinsic reward table uses $\mathcal{S} \times \mathcal{A}$ space; the value iteration table uses S space.
- Time complexity: There are $\mathcal{S} \times \mathcal{A} \times 2$ executions of the best response oracle. The complexity of the oracle depends on the time complexity of value iteration, which itself is not well understood. Each iteration of value iteration takes $O(|\mathcal{S}^2.\mathcal{A}|)$, but the number of iterations depends on the magnitude of the rewards, discount factor [50].

3.3 Intrinsic reward example

To demonstrate the effect of the intrinsic reward on the behavior of the agents, we consider four example scenarios of BRG agents playing IPD against naïve learners, shown in Table 1. Because the BRG agent observes both the regular episode return r and the intrinsic reward IR, it potentially has a gradient toward improvement in either metric. We observe that, depending on the current policies in the scenario, even when one of the signals is maximized, we can often still update the policy to improve on the other dimension.

The policies mentioned in this section are explained in detail in Sect. 4.

In situations where the environment reward is already maximized (Scenarios 1 and 2), the intrinsic reward allows the agent to make their policy more cooperative or robust; for example, in scenario 1, an `ALLD` (Always defect) BRG agent facing an `ALLD` opponent cannot improve their current reward, but improving the intrinsic reward leads to a `DTFT` (Tit-for-tat starting with defection)-like strategy, which makes cooperation possible in the future. Similarly, in scenario 2 an `ALLC` (Always cooperate) BRG agent facing a `TFT` (Tit-for-tat) opponent will discover its own vulnerability to defection through best responding to `ALLC` with `ALLD`. It can make itself less susceptible to future entrants by maximizing the intrinsic reward to reach a `TFT`-like policy.

In scenarios 3 and 4, the intrinsic reward is already maximized but the current reward may still be improved to make itself less exploitable to opponents. For example in scenario 3, the BRG agent is playing `TFT`, matched in

payoff by a best response ALLC. As such, the intrinsic reward suggests the opponent should become fully exploitable (resulting in reward T). However, the agent can still improve its current reward, by avoiding exploitation in the opening move at s_0.

4 Cooperation in repeated games

The basis for the IPD is the Prisoner's Dilemma (PD). Here, agents can choose from two actions: Cooperate or Defect; mutual cooperation is rewarded with a payoff R, and mutual defection is punished with a payoff P; a defector exploiting a cooperator will get a temptation payoff T, while a cooperator being exploited gets the sucker payoff S. This can be summarized using the payoff matrix $\begin{pmatrix} R & S \\ T & P \end{pmatrix}$, with $T > R > P > S$. Defection is the only dominant strategy and it is not Pareto efficient.

When the PD is repeated for an (uncertain) number of rounds, cooperation can be an equilibrium. In the IPD, the probability of having a next round is the continuation probability, δ. The IPD has infinitely many Nash equilibria if δ is large enough [51].

For example, a TFT (Tit-for-tat) strategy—cooperate in the first round and then follow the opponent's last action—can form the basis of an equilibrium profile (TFT, TFT). A TFT's threat of punishing defection encourages the opponent to cooperate. A pair of ALLD (Always defect) strategies also constitutes an equilibrium of the repeated game. Some other widely discussed strategies include ALLC (Always cooperate) and DTFT (Tit-for-tat starting with defection).

Players' strategies can use their memory of the N past rounds (memory-N) to choose actions. Following Press and Dyson [52], we restrict memory to one round, i.e., a player's action depends solely on the combined actions from the previous round, resulting in five possible states: (*S0—initial round*, *CC*, *CD*, *DC*, *DD*), where C denotes *Cooperation*, D denotes *Defection*. The actions are in the order of the focal player followed by the opponent.

When playing IPD in a population, pairs of agents are matched to play the game every round. In this paper, we use random matching. This matching procedure is common in EGT [30] and resembles a situation where agents are part of a large ecology and move through a series of interactions encoded as a game, which could itself consist of multiple steps, such as in the case of IPD. In population settings, the agents are usually kept anonymous and they make strategically independent decisions that shape the direction of the population in the long run.

4.1 Experiments

Here we describe and perform experiments to evaluate the impact our Intrinsic Reward has on convergence, stability, and robustness of efficient equilibria in the IPD. We design a number of experiments to evaluate the following hypotheses about our BRG agents; they

- *consistently* achieve cooperation, even against naïve agents;
- require *less computational resources* than an explicit opponent modeling strategy; and
- are *robust* to exploitation by new entrants.

Each of these hypotheses is tested in a separate experiment, which we describe in the following three subsections. Across all our IPD experiments, we use the same game structure, having payoffs $R = -1$, $S = -6$, $T = 0$, and $P = -5$ and a continuation probability $\delta = 0.95$.

Table 1 Scenario analysis for BRG agents. Here n is the expected duration of the game

Scenario	BRG	Opponent	Return r	IR
1	ALLD	ALLD	$P \cdot n$	$P \cdot n$
Expected behavior: BRG agents cannot maximize r, but IR can be improved to reach a maximum at DTFT. Maximizing IR makes cooperation possible for new entrants/in the future				
2	ALLC	TFT	$R \cdot n$	$P \cdot n$
Expected behavior: BRG agents cannot maximize r, but IR can be improved to reach a maximum at TFT. Maximizing IR makes BRG agent less exploitable for a new entrant/in the future				
3	TFT	ALLD	$S + P \cdot (n-1)$	$T \cdot n$
Expected behavior: BRG agents cannot maximize IR but r can be improved to reach a maximum at DTFT. Maximizing r makes the BRG agent less exploitable for the current opponent				
4	TFT	DTFT	$(T+S) \cdot \frac{n}{2}$	$T \cdot n$
Expected behavior: BRG agent cannot maximize IR but can attempt to improve r by cooperating more at state *CD*. This scenario also highlights the need of assuming a learning opponent				

4.1.1 Convergence of BRG agents

To evaluate if BRG agents consistently achieve cooperation, we conduct experiments where agents pair off against each other as BRG versus BRG, BRG versus naïve, and naïve versus naïve. Each pair is played to convergence multiple times from 100 different starting policies, to evaluate if convergence is robustly achieved. We sample the starting policies to choose the cooperate action with probabilities sampled uniformly, such that $\forall i, s : \pi_i(s) \sim \mathcal{U}(0.1, 0.9)$. We consider the achievement of cooperation when a pair converges to an *average* payoff above $(R + P)/2$, the average payoff of mutual defection and cooperation. This threshold implies that the agents are cooperating more than they are defecting.

Figure 1a presents the average learning trajectories of the three different pairs. As expected, we observe that a pair of BRG agents always achieves the reward for full cooperation, R. Even when playing against a naïve agent, the average reward achieved is close to full cooperation. We see this convergence behavior consistently, independent of the particular hyper-parameter setting used for both cases BRG vs BRG and BRG vs Naïve (Fig. 1b). For these and the following experiments, we therefore keep the hyperparameters constant. The learning rate is set to 0.01, with a batch size of 1000 and η of 0.5.

In Fig. 1a, we observe that the standard deviation is larger than the all BRG agents case. To investigate why, we look at the difference in payoff obtained by BRG and naïve agents, in Fig. 2a. We observe that BRG agents manages to exploit their naïve opponents to gain a reward that's slightly higher than R. BRG agents consistently achieve this stable payoff without a collapse to the low-payoff equilibrium *DD*. This stands in contrast with the experiments with a pair of naïve learners, which *typically* collapse to *DD* except under the occasional fortunate starting conditions.

We can further look into the characteristics of the resulting policies by looking at the distribution of their action probabilities in each state.

In Fig. 2b, the collective policy of agents in a particular state is depicted by a dot with each axis representing the corresponding player. A close examination of the figure reveals that BRG agents do not consistently commence with cooperation. However, they tend to compensate for instances of unilateral defection by subsequently cooperating (state DC depicted in red).

They control the naive agent to stay cooperative by threatening with a grim trigger-like period of defection (much higher chance of defection in CD or DD states where the opponent chose D action). Some level of cooperation is shown in *DD*, arguably to support recovery from accidental defection. Despite this, they manage to guide the opponents to full cooperation (under a wide range of η hyper-parameter settings, see Fig. 1b). We have further evaluated the algorithm's sensitivity to variations in learning rate and batch size. The corresponding figures are provided in Appendix A.

4.1.2 Computational complexity of BRG agents

To evaluate the computational benefit BRG agents bring to population games, we measure the wall-clock time it takes to perform one batch learning update *for the entire population* as the population size increases, across 100 runs per population size. We compare with the baseline of a naïve agent population, and with an opponent modeling strategy, LOLA+DiCE [6, 16]. LOLA agents achieve a high level of cooperation in IPD and the coin game by optimizing the policy against an assumed naïve learning step of the opponent.

We repeat the experiment for different population sizes and in every case, the agents play IPD with complete matching, meaning that every agent plays batch size number of games against every other agent in pairs. For LOLA implementation, we use the PyTorch version of LOLA-DiCE [53], for consistency with our own implementation. LOLA agents use their default parameters, except for an increased batch size of 1000, which we found was beneficial for stable convergence to cooperation. In our experiments, we only show LOLA with a single lookahead, which not only takes the least amount of time but also showed the best payoffs compared to 2 and 3 lookaheads.

Figure 3 shows how the performance of the algorithms changes with the population size. The top panel (a) shows that the convergence characteristics of the algorithms are unaffected by the size of the population; as observed, BRG and LOLA agent populations consistently achieve cooperation, while naïve agents tend to the low-payoff equilibrium.

However, looking at the wall-clock time each batch takes to process (Fig. 3b), we see that opponent modeling adds a significant runtime overhead, when compared against BRG agents. Furthermore, this overhead increases with population size, as agents have to train, model, and update policies separately for each opponent. In contrast, the overhead of increasing population size on naive or BRG agents is essentially constant, resulting in linear increase in the number of agents. Although we did not use multi-threading in our implementation, BRG agents could potentially update their policies in parallel resulting in minimal computational overhead compared with the baseline of naïve agents.

While it is not possible to characterize the run time of *all possible* OM techniques, in the standard case where an opponent model is kept for every agent the following

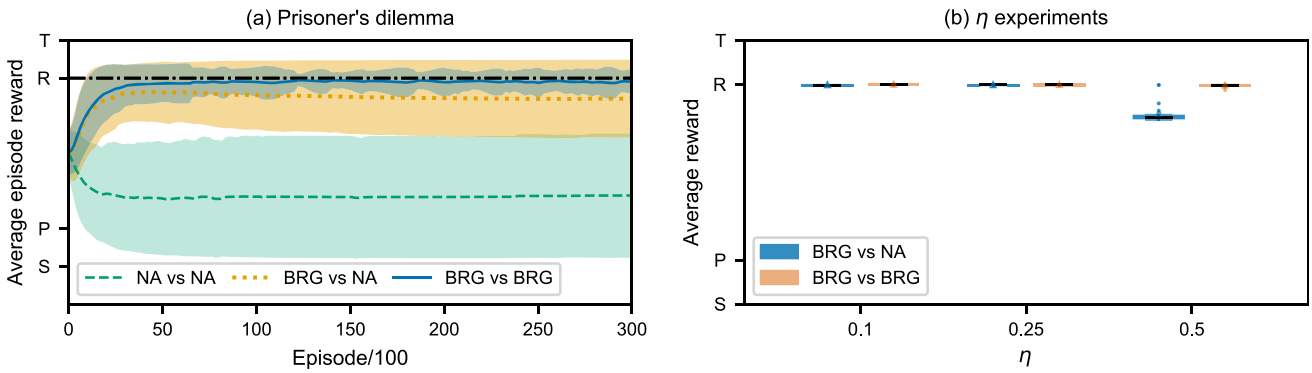

Fig. 1 Performance of different types of agents in the prisoner's dilemma. **a** Trajectories of average total episode rewards of both players. **b** Distributions of final average reward of episodes, as function of η. (NA: naïve agent. BRG: best response guided agent. x vs y: setting where agent type x plays type y one on one). The shaded area is one standard deviation above and below the average

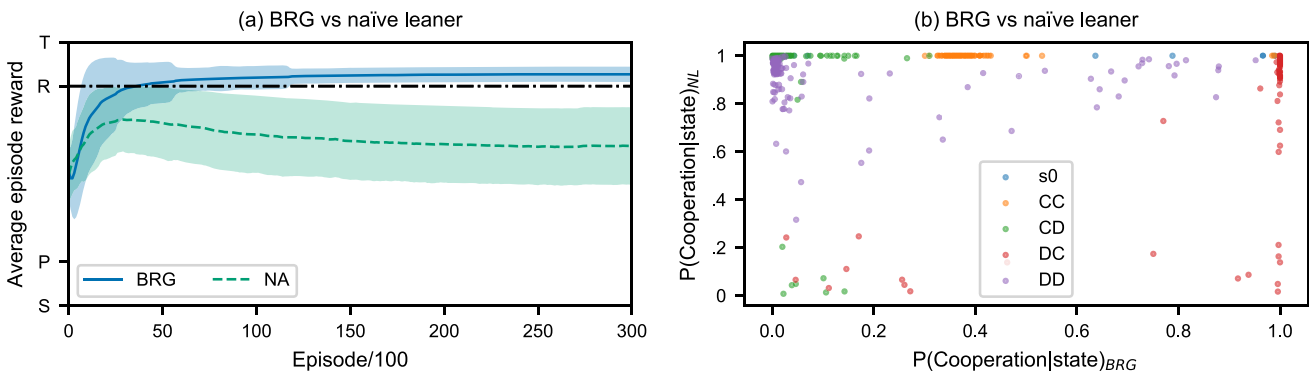

Fig. 2 Performance of BRG agents and naïve learners playing against each other. **a** Individual reward trajectories of BRGs and naïve learners when playing against each other (yellow in Fig. 1*a*), compared with full cooperation (black dashed line). **b** Scatter plot of the policies at the end of runs from panel *b*. (NA: naïve agent. BRG: Best response guided agent. x vs y: setting where agent type x plays type y one on one). Shaded area is one standard deviation above and below the average

intuition applies. The learning agent has to keep $n-1$ opponent models, which in the case of neural networks each individually is updated with batch size b/n. BRG keeps a single network, which is updated once with batch size b. Assuming an equal number of parameters in the nets, OM has to update n times the number of variables per gradient step.

In summary, BRG agents are successful in consistently achieving cooperation for different population sizes when compared with naive and OM approaches, as seen in Fig. 3a.

4.1.3 Exploitability and adaptability of BRG agents

Decentralized cooperation becomes essential when the imposition of a central controller is unfeasible. In such decentralized populations, it is imperative that agents demonstrate the capability to effectively compete with a diverse array of opponents and safeguard cooperation against potential disruptions introduced by new entrants. Here we conduct experiments to test the adaptability of BRG agents in such situations.

We evaluate the robustness of BRG agents against hostile ALLD (starting) policies, both from a cold start and against an invasion of new entrants. We perform four separate tests,

(a) both BRG and naïve agents starting from ALLD,
(b) new entrant naïve learners starting from ALLD,
(c) BRG agents facing fixed ALLD strategy from $t=0$, and
(d) new entrant fixed ALLD strategies introduced halfway.

In all cases, the ALLD strategies choose the defect action with probability 0.9 across all states, to allow for limited exploration of cooperation actions at all times. For these experiments, we consider populations of $n=20$ agents at the start, with 10 additional new entrants introduced halfway for conditions (b) and (d). Similar to previous experiments, we conduct each trial 100 times for statistical significance.

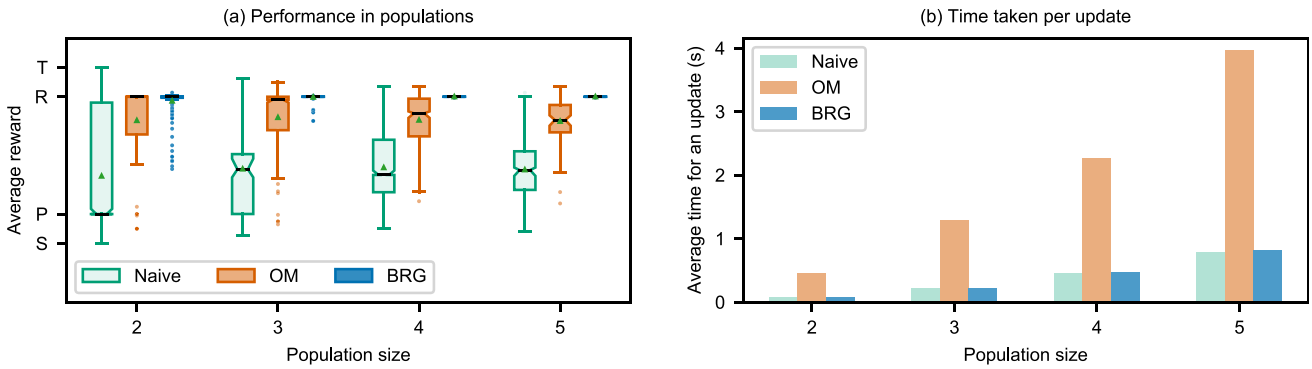

Fig. 3 Performance of Naïve, BRG, and OM agents as the (homogeneous) population increases from 2 to 5 agents, showing average of **a** step reward, **b** time per update step

The learning trajectories in each of these four cases are shown in Figs. 4 and 5. We observe that in all situations, self-aware agents are able to defend themselves against the hostile strategies. Starting from ALLD strategies, self-aware agents manage to bring the naïve agents to cooperation (Fig. 4a), without themselves becoming susceptible to exploitation against fixed policies (Fig. 5a).

Additionally, we observe that BRG are robust to hostile new entrants. Hostile naïve learners (Fig. 4b) are quickly converted to full cooperation, while against fixed ALLD new entrants, the BRG agents learn to resist exploitation successfully (Fig. 5b). We observe that in this setting, the ALLD agents initially manage to exploit the BRG agents some of the time, but as learning proceeds, the BRG agents gradually move to a full TFT strategy. This enables them to continue to cooperate among themselves, without losing much payoff against the fixed agents. With one-third of new fixed ALLD entrants, BRG agents still achieve a reasonable level of cooperation. They learn to cooperate only with other BRG agents securing themselves an average reward around $(2/3)R + (1/3)P$.

4.1.4 Other strategic scenarios

BRG agents do not explicitly account for the mixed-motive nature of the system and therefore do not categorize the environment. In order to assess their performance in both fully competitive and cooperative environments, we conduct additional experiments using a repeated coordination game and a zero-sum game. We do this to ensure that BRG agents are not disadvantaged in competitive games or games with structures that are not strictly about cooperation.

For the cooperative setting, we used a symmetric game where agents receive a reward of 5 for mutual cooperation, 1 for mutual defection and 2 otherwise Fig. 6. For a competitive environment, we used matching pennies Fig. 7. In both cases, we can see SA agents achieving the same equilibrium as naïve learners. This shows that this algorithm can perform even in situations where many different kinds of interactions arise.

5 Stochastic Prisoner's Dilemma

We now inspect how BRG agents perform in a stochastic game.

The Stochastic Iterated Prisoner Dilemma [17, SIPD] models a situation where synergistic cooperation results in increased overall social welfare. Importantly, this is a Markov game where the environment responds stochastically to the actions of the players.

Agents start off playing a low-payoff IPD with cooperation benefit b_1 and cooperation cost c and get the opportunity to play a game with higher benefits $b_2 > b_1$ only if both agents cooperate. This structure is illustrated in Fig. 8. A single defection will move the agents back to the state with a less rewarding game.

A SIPD agent with memory restricted to the last round observes 6 states; S_0 for $t = 0$ and CC_1, CC_2, CD_1, DC_1 and DD_1 based on the combined actions of the players in the last round and the game to be played next. S_0 refers to the initial state where previous actions are unknown and agents start with playing game1 with lower rewards. Thus, the memory-one strategy of an SIPD agent can be represented as $\pi = (\pi(\mathrm{C}|S_0), \pi(\mathrm{C}|CC_1), \pi(\mathrm{C}|CC_2), \pi(\mathrm{C}|CD_1), \pi(\mathrm{C}|DC_1), \pi(\mathrm{C}|DD_1))$ where $\pi(\mathrm{C}|S_0)$ is the probability of cooperation at state S_0.

We set up additional experiments using SIPD to evaluate,

- the robustness of BRG agents in a stochastic games, and
- the performance improvement compare to naïve learners.

Across all experiments, we use the following payoffs: $R_1 = b_1 - c = 3$, $R_2 = b_2 - c = 4$, $T_1 = b_1 = 4$,

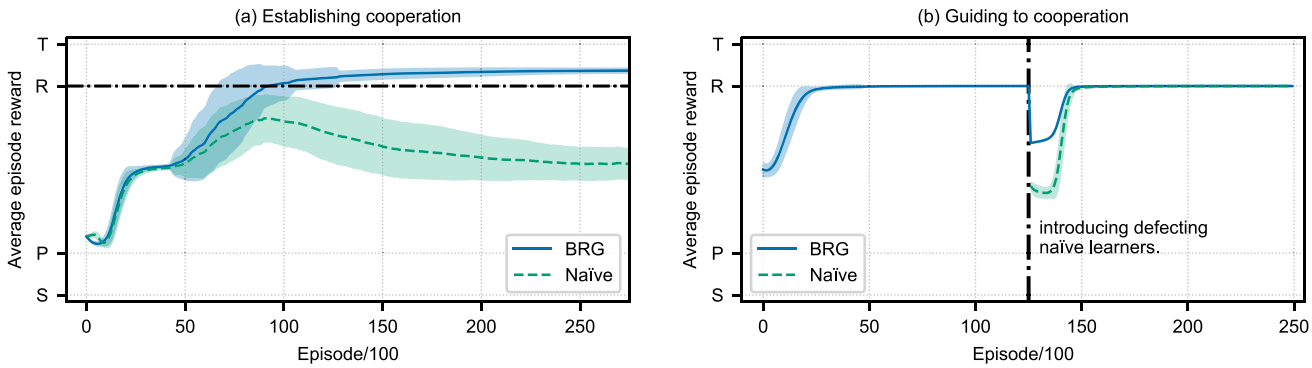

Fig. 4 Mean (line) and one standard deviation (shaded area) performance of BRG agents under adverse conditions: **a** starting from 90 percent defection against naïve learners, **b** defecting naïve learners as new entrants

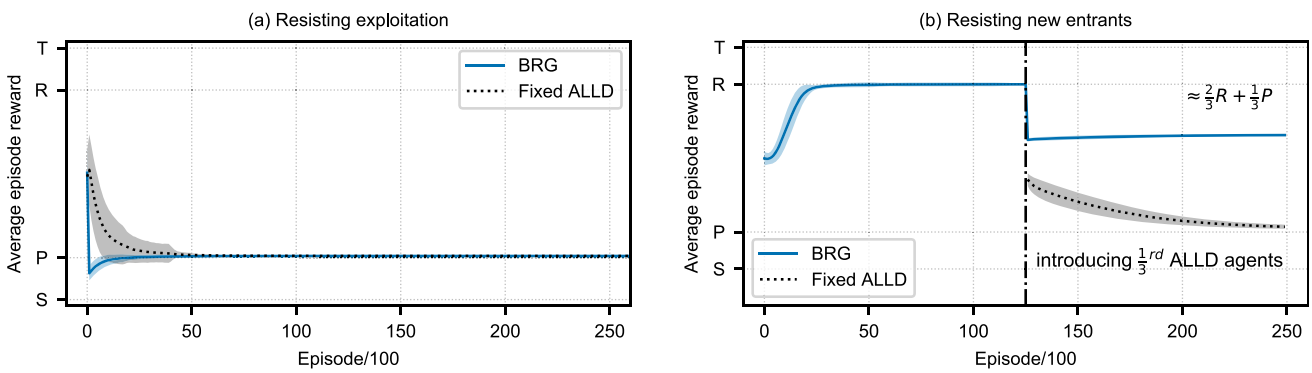

Fig. 5 Mean (line) and one standard deviation (shaded area) performance of BRG agents under adverse conditions: **a** learning against (fixed) ALLD agents, **b** fixed ALLD as new entrants

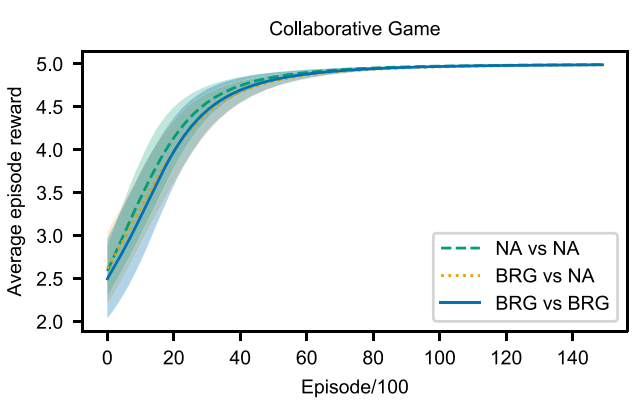

Fig. 6 Performance trajectories over average episode returns as both players learn in a collaborative game. (NA: naïve agent, BRG: best response guided agent)

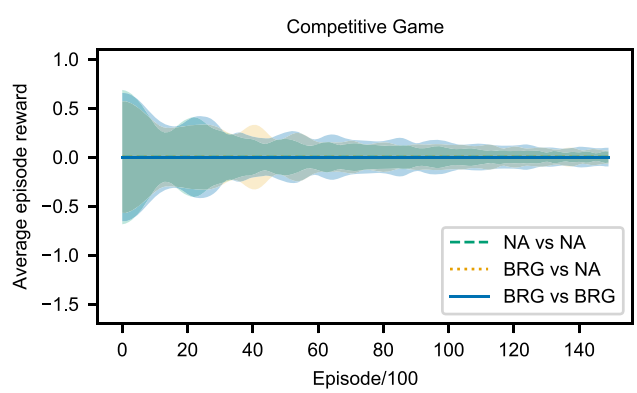

Fig. 7 Performance trajectories over average episode returns as both players learn in matching pennies zero-sum game (NA: naïve agent, BRG: best response guided agent)

$T_2 = b_2 = 5$, $S = -c = -1$, and $P = 0$ with 20 round for an episode, shifting agents to the better game with probability 0.95 upon mutual cooperation. Successful convergence to cooperation should result in a payoff close to $0.95R_2 + 0.05R_1$ ($\approx R_2$) as agents start each episode with the low pay-off game. The experiments here follow the same structure as those discussed before.

Figure 9a illustrates the reward trajectories of all three agent mixtures. BRG pairs achieve full cooperation with very low variance. BRG agents playing against naïve learners show a similar path.

As opposed to non-stochastic IPD, the experiments result in significantly lower variance as BRG agents no longer benefit from attempts to exploit naïve learners

through occasional defection. Any such attempts are more likely to shift the game to the low pay-off reward structure. Figure 9b further verifies this with high levels of cooperation.

For our experiments, we used a learning rate of 0.01 and a batch size of 1000. η is set to 0.25 when playing against BRG agents and 0.1 when playing against naïve learners—in a similar pattern to the deterministic game the choice of η does not have dramatic effects.

In summary, BRG agents perform well in the stochastic version of the game: they establish and maintain cooperation.

6 Cooperation in complex policies

Despite the strategic complexity associated with the IPD—infinite equilibria with uncountably infinite strategies—it is a simple two-action simultaneous game where actions directly map to the features of cooperation or defection. Mixed-motive scenarios can be much more environmentally complex. In this section, we assess the performance of BRG agents in an environment where cooperation or defection is not a straightforward one-time decision but rather the outcome of a sequence of actions taken across multiple time steps.

In the coin game introduced by Lerer and Peysakhovich [18], fostering cooperation requires agents to limit resource collection, an action that would otherwise diminish their opponents' rewards. This makes cooperation a combination of navigation decisions across multiple time steps, making cooperation a feature of the policy. Establishing cooperation becomes more problematic for naive reinforcement learners because they have to learn the dynamics of the action in addition to the semantics of collaboration. The coin game presents larger state–action spaces when compared to the IPD, increasing computational complexity. But, this complexity is representative of real-world cooperation dilemmas. Due to this property, this game has been used extensively in the literature to gauge performance of algorithms in complex cooperative settings [6, 18, 35].

In the coin game, two agents (red and blue) are placed on a grid where (red and blue) coins appear uniformly randomly. Agents receive a reward of 1 for picking up any

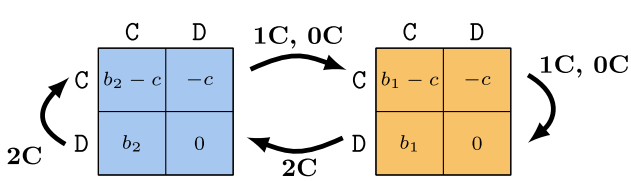

Fig. 8 Stochastic iterated prisoner's dilemma [17]

coin, but lose 2 points if the opponent picks a coin of the focal agent's color. If both agents reach the coin simultaneously, the opponent receives 1 and the agent of the same color receives -1. A socially optimal behavior entails agents coordinating on the color of the current coin to avoid losing points, but picking up any coin is always individually dominant. This results in naïve learners greedily picking up coins making the average reward zero. A standard situation in this game is depicted in Fig. 10. In both cases, a socially optimal result entails the agent closest to the coin waiting for a coin of their color.

Lerer and Peysakhovich [18] introduce the coin game to test amTFT, an algorithm that allows agents to shift between cooperative and safe policies. Foerster et al. [6] use a simpler 3×3 grid to assess `LOLA` agents with and without opponent policy information. They gain little success with modeled opponents but achieve better results when perfect opponent information is provided. Raileanu et al. [54] use a more complex fully collaborative version, with 3 players and coins of 3 different colors in a 8×8 grid to test another opponent modeling algorithm, SOM (self-other model). More recent work of Lu et al. [35] evaluates the performance of Model Free Opponent Shaping (MFOS) in a 3×3 grid similar to [6] and report a significant performance improvement. Badjatiya et al. [36] show agents can achieve near full cooperation with Status-Quo Learners that assume a stationary environment in environments with two agents.

We evaluate the coin game restricting the memory of agents to each player's last action. A state of the game is represented by a vector (d_f, d_o, l, h_f, h_o), where d_f is the distance (number of grid squares) from the focal agent to the coin, d_o is the distance from the opponent to the coin, l is the coin color, h_f is the focal agent's cooperation history, and h_o is the opponent's cooperation history. To interpret each other's move, agents use the following heuristic: moving toward a coin of the opponent's color is perceived as defection and moving away from a coin of different color is perceived as cooperation. Distance to the coin is defined as the minimum number of steps required to reach the coin, and cooperation history can be -1, 1 or 0, depending on whether the agent has defected, cooperated, or unknown (at the beginning). We use a $3x3$ grid where a new coin is always generated 2 steps away from both agents at all times. This simplified version facilitates the intrinsic reward calculation for the BRG agents.

For experiments, we used policy gradient with learning rate 0.001, batch size 20, and 100-step episodes. To ensure continuous exploration, we clip policies to $\pi(s,a) \in (0.05, 0.95)$. Similar to IPD and SIPD, we use value iteration for intrinsic reward calculation, but gradually anneal η from $0 \rightarrow 0.5$. Setting $\eta = 0$ boosts

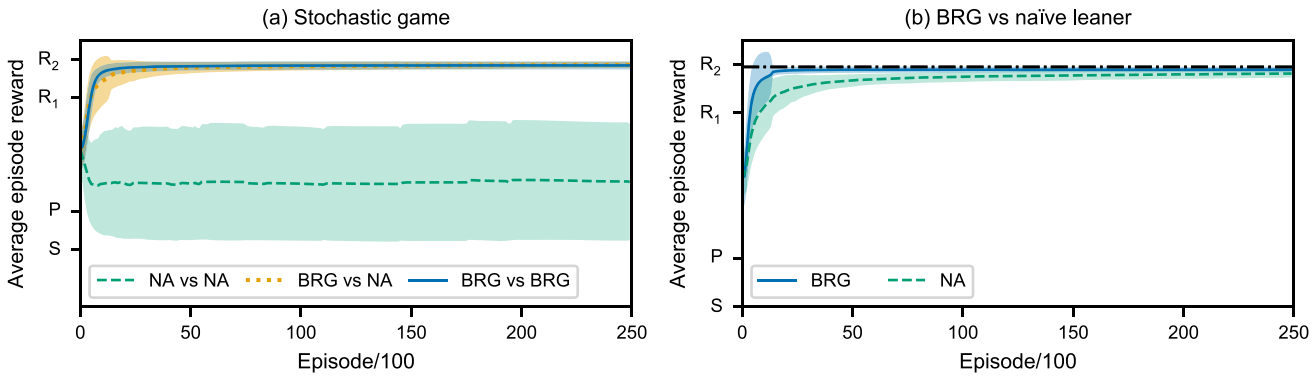

Fig. 9 Performance of different types of agents in the stochastic prisoner's dilemma. **a** Trajectories of average total episode rewards of both players. **b** The separate individual trajectories of BRGs and naïve learners when playing against each other (i.e., BRG vs NA in (**a**))

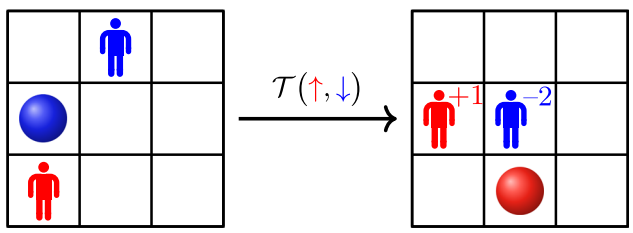

Fig. 10 Coin game: socially optimal behavior entails agents coordinating on the color to avoid losing points, but picking up any coin is individually dominant

exploration to states with higher future rewards, slowly correcting back to $\eta = 0.5$ to encourage exploitation.

Results are presented for episodes with 100 steps. We performed experiments for scenarios, naïve against naïve, and BRG against BRG, 20 times each to generate Fig. 11a. As expected, naïve learners converge to the greedy strategy where each agent attempts to pick all coins receiving an average reward of zero. BRG agents achieve a significant level of cooperation reaching an average reward of 10 where the reward for full cooperation is 25. For comparison purposes, we use the ratio, (*collected coins of own colour/total coins picked*), Fig. 11b, similar to [6] and [35].

Although a direct comparison of results is not meaningful with prior work, we can qualitatively compare the results for the mixed-motive coin game. BRG agents achieve a $(\textit{own colour}/\textit{total coins picked})$ ratio of around 0.65 which is on par with LOLA agents, PPO [55] agents playing against MFOS achieve around 0.55, and SQLearners achieve around 0.98 [36]. Based on these results, we see that BRG agents are comparable with existing approaches. We have also identified that provided more computational resources BRG agents could improve their performance, particularly by introducing more elaborate steps in recognizing cooperation which we currently do with a very simple heuristic.

Due to the complexity of best response to best response calculation for two agents each having 72 (2 colored coins$\times 3 \times 3$ grid size $\times 4$ actions), we compressed the state action space. Calculating this optimal Q[s,a] table takes considerably longer for each gradient update step, versus the calculation for the compressed model. This heuristic is not essential, and given resources, a full calculation or a different heuristic may be used (see e.g., [36]).

7 Conclusions and future work

Autonomous agents learning to maximize collective rewards will need to be robust to both the temptation to cheat others and the risk of other agents attempting to exploit them. While such concerns can be addressed by modeling opponents explicitly when the population size is small and agents are individually identifiable, this strategy becomes intractable when the population is large and interactions are sparse. This situation is important. Consider the case of a self-driving car interacting with others. The agent will face a series of interactions, some of which have a cooperative structure, and some of which are more coordination or pure competition. At the same time, each interaction entails a new opponent. This kind of setup can also arise for example in energy markets, where depending on changes in the environment the structure of incentives can switch from cooperation-like, to fully competitive. All of this across large groups of agents makes opponent modeling impractical.

We present an effective approach to circumvent this problem, by augmenting an agent's internal reward signal with a best response guided component derived from the best response to the agent's current policy, i.e., forgoing all

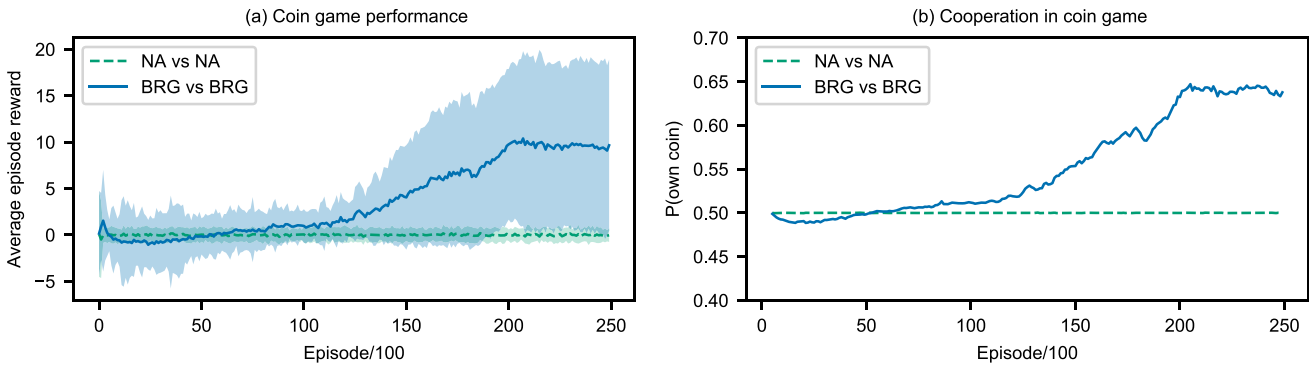

Fig. 11 Naïve learners and BRGs playing the coin game. **a** Trajectories of average total episode rewards of both players. **b** Percentage of own color coins collected

information from opponents. We demonstrate that this reward signal allows our agents to consistently establish cooperation with naïve learning agents in an iterated prisoner's dilemma and a stochastic game with prisoner's dilemma-like reward structure. The resulting policies are robust to new entrants and shown to be significantly more scalable than opponent modeling approaches as the population grows. These results hold for a simple IPD game, but also for a stochastic game featuring different states.

In addition, the algorithm performs equally well as naïve learning agents in fully competitive and cooperative environments. In games with higher state spaces like the coin game, BRG agents achieve a significant level of cooperation compared to naïve learners that converge to the greedy approach.

Future work should focus on applying our reward structure to games and environments with more complex states. Examples include more explicit scenarios, for example, self-driving cars, where every intersection is akin to its own game but actions play out temporally, and smart energy solutions like voluntary load shedding to avoid blackouts and curtailment of solar panels under voltage constraints. Such settings will also motivate methods other than optimal best response to calculate the intrinsic reward, instead learning the signal in a separate self-play reinforcement loop.

The intrinsic reward calculation does not necessarily have to be value iteration. One proven method for best response calculation is self-play (add citations). Although it can be computationally expensive, it could suit situations where the agents do not have access to the reward structure or transition probabilities but can interact with the environment via a simulator. Another method that can be explored is using value networks to learn to predict value estimates over time (as opposed to having access to an optimal solution oracle). Care would need to be taken to handle the drift in value that would occur as a result of the agent's policy changing over time, something which may be addressed via two time-scale training. These approaches may improve the applicability of our framework to larger state spaces and unknown world dynamics, however at the cost of significantly longer computational time to calculate intrinsic rewards, and potentially sub-optimal estimates. Future work will need to evaluate this trade-off.

A Appendix

A.1 Sensitivity to hyperparameters

In this section, we examine the sensitivity of BRG agents to the hyperparameters of learning rate and batch size when interacting with both similar agents and naive learners. This analysis looks at the final average reward of both players and its variance for 15 pairs of agents after 3000 updates.

As illustrated in Fig. 12a, BRG agents playing against other BRG agents consistently achieve full cooperation, regardless of the learning rate. However, the average score of agents improves with a high learning rate of 0.1, whereas a learning rate of 0.01 results in significant variance. This finding aligns with the observations in Fig. 2a, where BRG agents learn to exploit naive learners, achieving a reward higher than that of pure cooperation. This exploitation leads to higher variance in the rewards of naive learners when playing against BRG agents, as depicted in Fig. 2a, and is reflected in the combined average rewards.

In Fig. 12b, it is shown that BRG agents consistently achieve full cooperation regardless of the batch size.

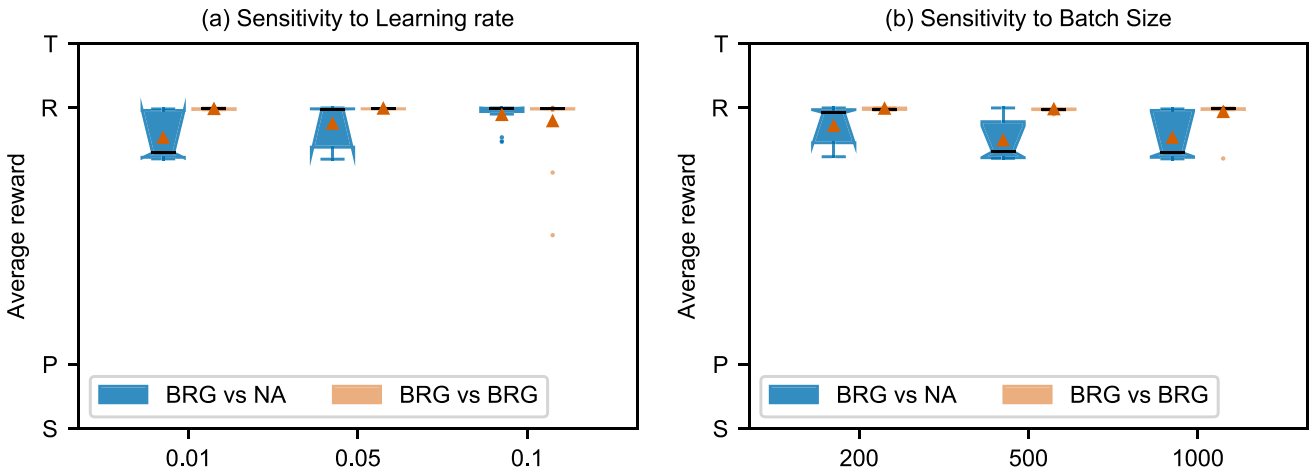

Fig. 12 Distributions of final average reward of episodes, as functions of learning rate and batch size. (NA: naïve agent. BRG: best response guided agent. x vs y: setting where agent type x plays type y one on one)

However, when interacting with naive learners, a higher batch size leads to a lower average reward for the players. This observation aligns with Fig. 2a, where BRG agents demonstrate an ability to exploit naive learners.

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Data availability The data and the code used to support these findings of this study are available upon request and on publication.

Declarations

Conflict of interest All authors have no conflict of interest.

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